

Midterm 1 practice test

Prepared October 1, 2026 for the October 2 midterm

Name: _____ Date: _____

Suggested time: 75 minutes for Questions 1-4. Optional bonus: 20 minutes.

Practice marking: 40 main marks + 10 bonus marks. These times and marks are for this mock test only.

Show your diagrams, equations, substitutions and units. State directions and sign conventions. Attempt the questions before opening the separate answer key. Take short breaks if using this as a study session.

Why these questions?

The instructor's September 29 announcement specifies equilibrium, electric flux / enclosed charge, one of four field derivations, and maximizing a supplied field function. The bonuses concern setting up and solving a non-uniform-charge integral. This mock follows those categories; it does not predict the actual questions.

Practice formula sheet

$$k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2; \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$$

$$g = 9.80 \text{ m/s}^2; \quad e = 1.602 \times 10^{-19} \text{ C}$$

$$F = k |q_1 q_2| / r^2; \quad \mathbf{F} = q\mathbf{E}; \quad \text{equilibrium: sum of forces} = 0$$

Uniform-field flux: $\Phi = EA \cos(\theta)$, with θ measured from the surface normal.

$$\text{Closed-surface flux: } \Phi_{\text{net}} = Q_{\text{enclosed}} / \epsilon_0.$$

$$dq = \lambda dl \text{ for a line; } dq = \sigma dA \text{ for a surface; } dE = k dq / r^2.$$

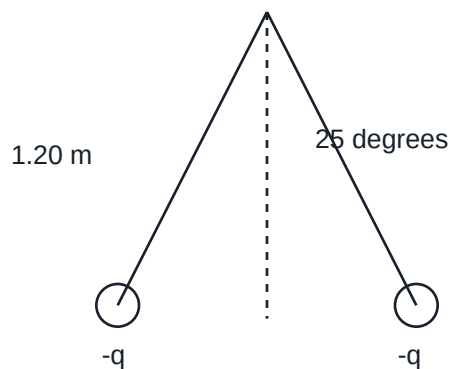
This is a study aid, not a transcription of the instructor's supplied formula page.

1 Charged spheres in equilibrium

10 marks | Suggested time: 20 minutes | Adapted from homework 2, textbook 21.63

Two identical spheres, each of mass 15.0 g, hang from a common point on strings of length 1.20 m. Both have the same negative charge. At equilibrium, each string makes 25.0 degrees with the vertical.

- (a) Draw a free-body diagram for one sphere. [2]
(b) Find the separation of the spheres and the charge on each sphere, including its sign. [6]
(c) Without calculation, predict how the angle changes if both strings are shortened while the charges remain unchanged. Explain. [2]



Working / free-body diagram:

2 Electric flux and enclosed charge

10 marks | Suggested time: 15 minutes | Adapted from homework 3, textbook 22.8 and 22.6

Three point charges are $q_1 = +4.00 \text{ nC}$, $q_2 = -7.80 \text{ nC}$ and $q_3 = +2.40 \text{ nC}$. Positive flux is outward.

(a) Find the net flux through each closed surface: S_A encloses only q_1 ; S_B encloses q_1 and q_2 ; S_C encloses all three. [3]

(b) If q_3 moves closer to S_B but stays outside it, does the net flux through S_B change? Explain. [2]

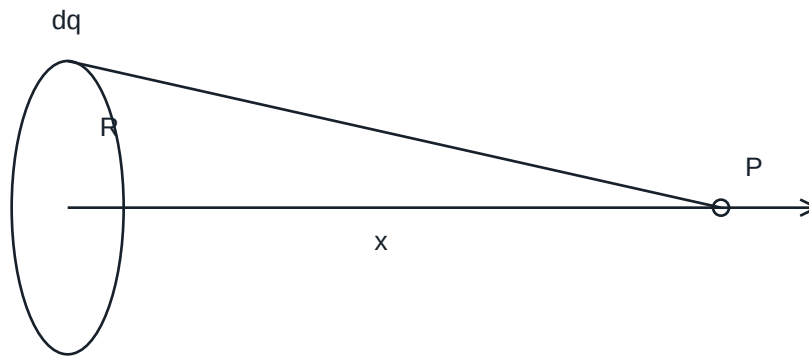
(c) A cube of side 0.100 m is placed in a uniform field of magnitude 4.00×10^3 N/C, directed 53.1 degrees from +x toward +y. Find the flux through each of the six faces (label them +x, -x, +y, -y, +z, -z), the total flux, and the net enclosed charge. [5]

3 Derive an electric field

10 marks | Suggested time: 25 minutes | Ring case from the D2L derivation handout

A thin ring of radius R carries a uniformly distributed positive charge Q . Point P is on its axis, a distance $x > 0$ from the centre. Derive the electric field at P from Coulomb's law.

- Define a charge element dq and its distance to P. [2]
- Explain which components cancel and which add. [2]
- Set up and evaluate the integral for the axial field. [4]
- Check the limits at $x = 0$ and at x much larger than R . [2]



Derivation:

Bonus Non-uniform charged rod

10 bonus marks | Suggested time: 20 minutes | Original variant of posted practice midterm Q4

A thin rod lies on the x-axis from $x = 0$ to $x = L$. Its linear charge density is $\lambda(x) = \lambda_0(1 + x/L)$, with $\lambda_0 > 0$. Point P lies on the same axis at $x = -d$, where $d > 0$.



Bonus 1: Set up an integral for the x-component of the field at P. State dq , the source-to-point distance, the limits and the direction. Do not evaluate it yet. [5]

Bonus 2: Evaluate your integral. Then calculate the field for $\lambda_0 = 2.00 \text{ nC/m}$, $L = 0.500 \text{ m}$ and $d = 0.100 \text{ m}$. [5]

After the timed practice

Optional consolidation | Not included in the 40-mark mock test

The real test may ask for any one of the four derivations. On fresh paper, also reconstruct:

1. Dipole: field on the perpendicular bisector, including direction and far-field limit.
2. Uniform rod: field on the rod's axis beyond one end (homework 2, 21.79).
3. Uniform disk: build it from concentric rings and integrate.

Extra questions, if needed

Textbook 21.65 (HW2): equilibrium of a negatively charged ball in a horizontal field.

Textbook 22.2 (HW3): distinguish the angle to a plane from the angle to its normal.

Posted 2004 practice midterm Q3: equilibrium and the number of transferred electrons.

Posted 2004 practice midterm Q4: non-uniform rod integration.

Source links

[Instructor hints, September 29](#)

[Four field derivations](#)

[Posted 2004 practice midterm](#)

[HW2 questions and worked solutions](#)

[HW3 questions and worked solutions](#)

Question numbering refers to the course's assigned textbook problems. Prompts here are paraphrased or adapted. The bonus geometry is explicitly defined for this mock and should not be assumed identical to the old practice test.